

Title Page

Title: Personalisation in Nutrition & Health Applications

Description: Analysis of how leading health apps tailor user experience based on user goals, preferences, and dietary needs

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1. Introduction

Personalisation is becoming more and more important in health and nutrition applications to enhance user engagement and results. These apps offer a customised experience by modifying recommendations according to user objectives, dietary choices, and behavioural patterns. This study examines the implementation of personalisation elements in three applications: Fastic, MyFitnessPal, and NutraCheck.

2. Methodology

- Three apps from market of same industry were used: Fastic, MyFitnessPal, NutraCheck
 - Screenshots were collected from each of three apps and their features like onboarding process, goal setting, recommendations, alert and dietary handling were observed.
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3. App Analysis

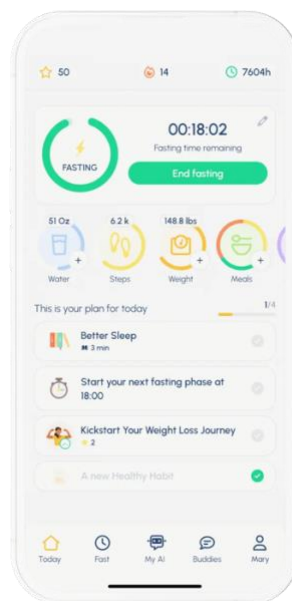
3.1 Fastic

Fastic focuses on intermittent fasting and lifestyle habits.

Personalisation Features:

- Users choose objectives (health improvement, weight loss).
- Personalised fasting times (e.g., 16:8, 18:6)
- Reminders and daily progress monitoring
- AI coach recommendations based on fasting habits

Screenshot Analysis:



- **Insight:**
The user's fasting timer, daily progress, and recommended actions including staying hydrated, getting better sleep, and beginning the next fasting phase are all displayed on this screen. This demonstrates behaviour-based customisation, in which the application customises the user experience based on a chosen fasting schedule. It offers customised reminders and habit recommendations based on fasting progress in place of comprehensive nutrition tracking, promoting consistency and lifestyle modifications.

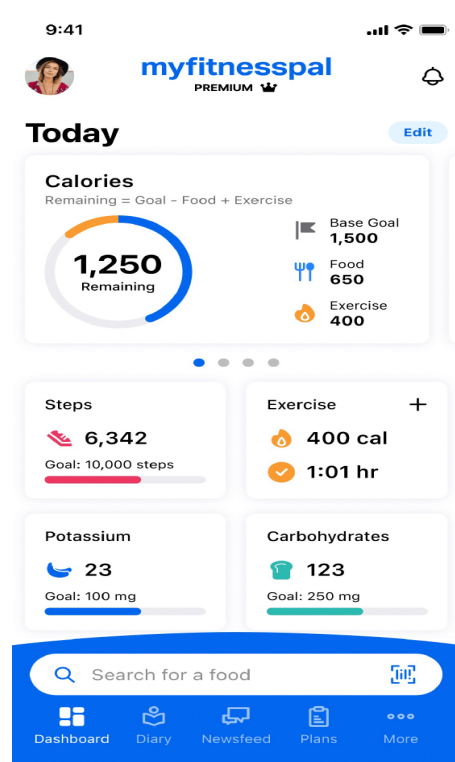
3.2 MyFitnessPal

A comprehensive calorie tracking and nutrition app.

Personalisation Features:

- Calorie targets according on height, weight, and degree of activity
- Monitoring macronutrients (protein, carbohydrates, and fat)
- History-based dietary suggestions
- Integration with fitness data through the display of exercises, steps, etc.

Screenshot Analysis:



Insight:

The user's daily calorie summary, steps, activity, and nutrient consumption are all displayed on this screen. For ease of tracking, the data is divided into distinct areas. This illustrates data-driven customisation, in which user inputs such as weight and activity level are used to modify the calorie objectives (1,500). The app shows how suggestions are constantly adjusted to user behaviour by dynamically updating remaining calories based on food intake and exercise.

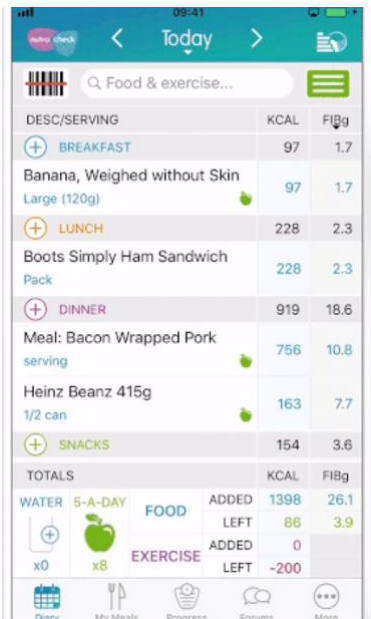
3.3 NutraCheck

Focused on calorie counting with UK/AU food databases.

Personalisation Features:

- Objective-based weight and calorie goals
- The traffic light food system (indicators of health and unhealth)
- Tailored meal recommendations
- Dashboards for tracking progress

Screenshot Analysis:



Insight:

NutraCheck uses clear dashboard to show all the meals along with amount and calorie for better understanding. All meals that need to be taken according to diet goals and plans pops on the screen and can also get details in meals section and also the exercises and workout plans is also available for pre and post workout. The progress is also visible in the progress tab which displays total calorie intake of day and workout plan as well

4. Personalisation Based on User Goals

App	Goal Input	How It Affects Suggestions
Fastic	Fasting goal	Adjusts fasting schedule & reminders
MyFitnessPal	Weight & fitness goal	Sets calorie + macro targets
NutraCheck	Weight goal	Recommends food choices & intake limits

Key Finding:

User goals directly influence:

- Daily targets
- Notifications

- Recommended actions

5. Dietary Restrictions & Allergy Handling

App	Goal Input	How It Affects Suggestions
Fastic	Fasting goal	Adjusts fasting schedule & reminders
MyFitnessPal	Weight & fitness goal	Sets calorie + macro targets
NutraCheck	Weight goal	Recommends food choices & intake limits

Analysis:

- MyFitnessPal offers most comprehensive control
- NutraCheck emphasises making decisions more easily.
- Fastic mostly disregards specific dietary requirements.

6. Comparison of Recommendation Systems

Feature	Fastic	MyFitnessPal	NutraCheck
Recommendation Type	Behaviour-based	Data-driven	Visual guidance
Complexity	Low	High	Medium
Adaptability	Moderate	High	Moderate

Observation:

- MyFitnessPal = most advanced but complex
- Fastic = simplest but limited
- NutraCheck = balanced usability

7. Key Insights & Patterns

The investigation reveals that the information gathered during onboarding, including user goals, weight, activity level, and nutritional preferences, is what drives personalisation in all three programmes. These initial inputs serve as the basis for how each app customises its notifications, recommendations, and design. As user behavior—such as logged meals, regular fasting, or exercise levels—is continuously monitored and incorporated into the system, this personalisation gets more sophisticated over time. The way these apps handle customisation shows a distinct trend. Instead of providing in-depth nutritional analyses, Fastic focuses on behavior-based customisation, helping users through fasting routines and habits. MyFitnessPal takes a data-driven approach, producing accurate calorie and macronutrient recommendations through substantial user input and tracking. NutraCheck, on the other hand, is a visual-guided approach that makes decision-making easier by using clear signs like color-coded meal evaluations. This illustrates a trade-off between

usability and complexity: simpler visual systems improve accessibility and ease of use, while data-intensive systems offer more precision and control.

8. Conclusion

In summary, the three applications show different methods of customisation in relation to health and nutrition management. Fastic is appropriate for users who are focused on fasting and lifestyle changes because it places a high priority on habit building and behavioural assistance. For customers who need exact tracking and control over their food consumption, MyFitnessPal provides an extremely analytical and detailed experience. By combining user-friendly visual signals with limited personalisation, NutraCheck offers a balanced approach that makes it accessible to a wider audience. In the end, the user's tastes and objectives determine how effective personalisation is. Apps like Fastic or NutraCheck may be more beneficial to users who want simplicity and convenience of use, whereas MyFitnessPal is a better option for people who need in-depth information and customising.

9. References (APA Style)

You can paste this directly:

- Fastic GmbH. (2023). *Fastic fasting app*.
- MyFitnessPal Inc. (2023). *MyFitnessPal application*.
- NutraTech Ltd. (2023). *NutraCheck calorie counter*.
- Teixeira, P. J., et al. (2015). Motivation and behaviour change in health apps. *Health Psychology Review*, 9(1), 1–15.
- Direito, A., et al. (2017). Do physical activity and dietary smartphone apps incorporate behaviour change techniques? *BMC Public Health*, 17(1).